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Data and Work Sheets - Print or bring it on an electronic device

Work and Energy - Physics 1401A
2026-2027

Please circle the appropriate values											
Course	**1101A**		**1201A**		**1401A**		**1501A**				
Lab Section	**002**	**003**	**004**	**005**	**006**	**007**	**008**	**009**	**010**	**013**	**014**
Lab Subsection	**A**	**B**	**C**	**D**							
Name	**First:** Beckett					**Last:** Graham					
Student #	2	5	1	5	5	0	6	5	6		
Lab Partner	**First:**					**Last:**					
Lab Station #											
Date	oct 20										
Demonstrator	Dennis + Raman										

Disclaimer: Please note that some but not all questions in this lab writeup will be graded.

EXPERIMENT 1: PRELIMINARY ADJUSTMENT OF THE APPARATUS

DATA AND CALCULATIONS

Levelling the air track

Place the car at one end of the track and send it through both photogates. If the track is horizontal, the timers should read the same time to within 5%. If necessary, adjust the leg(s) at one end of the track in order to level it. (Adjust at the end that is not connected to the air hose.)

Timer reading: $t_1 =$ 0.403 $t_2 =$ 0.397

Percent difference between the two readings: (Should be less than 5% for better accuracy.)

$$P_{diff} = \frac{|t_1 - t_2|}{(t_1 + t_2)/2} \times 100\% = \frac{0.006}{0.400} \times 100\% = 1.5\%$$

What conclusion can you draw from the percent difference test?

the tracks is Almost perfectly Level

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EXPERIMENT 2: CONSERVATION OF MECHANICAL ENERGY

Units in data: Measure x_1, x_2, ℓ, D and H in centimeters (cm) and record in cm. Convert data into SI units in the first step in your calculation. All calculations must be done in SI units.

Uncertainty in x_1, x_2: These are 'positions' of the photogates, and the uncertainty is $\pm$ 0.05 cm.
Uncertainty in ℓ, D, H and L: These are length measurements. The uncertainty is $\pm$ 0.1 cm. However, estimate δD to be higher since D is difficult to measure.
Uncertainty in m: Add the uncertainties in masses and the pan balance used.

Record absolute uncertainties in data to one significant figure. Follow the rules for subtraction, addition, multiplication and division. Keep the correct number of significant figures in derived quantities.

DATA

Table 1: Dimensions of the Apparatus

Item	Measurement & Uncertainty
1st Photogate	$x_1 \pm \delta x_1$ = 74.0 ± 0.05cm
2nd Photogate	$x_2 \pm \delta x_2$ = 129.0 ± 0.05cm
Length of flag on air car	$\ell \pm \delta \ell$ = 12.1 ± 0.1 cm
Distance between track legs	$D \pm \delta D$ = 100.0 ± 0.3cm
Mass of air car	$m \pm \delta m$ = 217.9g ± 0.1g
Height of aluminum tube	$H \pm \delta H$ = 2.5cm ± 0.1cm = 2.5 cm ± 0.1 cm = 2.6 cm ± 0.1cm

CALCULATIONS

Convert all units into SI units in your calculations.

δH_{avg} = meter ruler reading uncertainty + $^1/_2$ (H_{max} - H_{min})

$\delta L = \delta x_1 + \delta x2$

Distance between photogates **L = x_2 − x_1**	$L \pm \delta L$ = 1.29 − 0.74 ± (0.05+0.05m) = 0.55m ± 0.1 m
Average H	$H_{avg} \pm \delta H_{avg}$ = 0.0253 ± 0.001m

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Calculate the vertical height h between the points A and B. Find the absolute uncertainty δh. Write your answer in the form $h \pm \delta h$ units. Since h is an intermediate value, keep δh to two significant figures (Notes to Students: Section 3.4).

$$h = L \sin\theta = \frac{LH_{avg}}{D} = \frac{(0.55)(0.0253)}{1} = 0.0139\,m \tag{14}$$

$$\frac{\delta h}{h} = \frac{\delta L}{L} + \frac{\delta H_{avg}}{H_{avg}} + \frac{\delta D}{D} = \frac{0.1}{0.55} + \frac{0.001}{0.0253} + \frac{0.003}{1} = 0.224$$

$$\delta h = 0.224 \times 0.0139 = 0.00311\,m$$

Record $h \pm \delta h$ as an intermediate value:

$$h \pm \delta h = 0.0139 \pm 0.00311\,m$$

(Intermediate value of δh has two significant figures.)

Calculate the car's initial gravitational potential energy at A, $U_A = mgh$. Use $g = 9.804 \pm 0.001$ m/s^2. Then find the fractional ($\delta U_A / U_A$) and absolute uncertainties (δU_A) in U_A. Record U_A in the form $U_A \pm \delta U_A$ units.

$$U_A = m\,g\,h$$
$$= (0.2174)(9.804)(0.01390)$$
$$= 0.02969\,J \tag{15}$$

$$\frac{\delta U_A}{U_A} = \frac{\delta m}{m} + \frac{\delta g}{g} + \frac{\delta h}{h}$$
$$= \frac{0.0001}{0.2174} + \frac{0.001}{9.804} + \frac{0.00311}{0.01390}$$
$$= 0.00666\,J$$

Intermediate value: $U_A \pm \delta U_A = 0.027 \pm 0.007\,J$

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The values in the last column E_A - E_B must be close to zero. If they are not, your data may have a problem. Verify your data and calculations, and contact the demonstrator if necessary.

Table 2: Energy of the Air Car

(initial) t_A (s) ± 0.001	(final) t_B (s) ± 0.001	v_A (m/s)	v_B (m/s)	K_A (Joules)	(total initial energy) $E_A = U_A + K_A$ (Joules)	(total final energy) $E_B = K_B$ (Joules)	E_A - E_B (Joules)
0.222	0.160	0.563	0.781	0.0345	0.0642	0.0665	−0.00230
0.222	0.160	0.563	0.781	0.0345	0.0642	0.665	−0.00230
0.222	0.160	0.563	0.781	0.0345	0.0642	0.665	−0.00230
0.221	0.160	0.566	0.781	0.0349	0.0646	0.0665	−0.00190

Sample calculation of E_A - E_B

Select one trial from Table 2, and starting from the first column, show how you will calculate v_A, v_B, K_A, K_B, E_A and E_A - E_B. Show the equation and the first step of substituting numerical values into the equation and units. Each t value in the first two columns has three significant figures. Carefully think of how many significant figures you should keep in subsequent columns.

$v_A = l/t_A$

$= 0.125\text{m}/0.222\text{s} = 0.563\text{m/s}$

$v_B = l/t_B$

$= 0.125\text{m}/0.160\text{s} = 0.781\text{ m/s}$

$K_A = {}^1/_2\, mv_A^2$

$= \frac{1}{2}(0.217\text{kg})(0.563)^2 = 0.0345\text{ J}$

$K_B = {}^1/_2\, mv_B^2$

$= \frac{1}{2}(0.217\text{kg})(0.781)^2 = 0.0665\text{ J}$

$E_A = U_A + K_A$

$= 0.02969 + 0.03454 = 0.0642\text{ J}$

$E_B = K_B$

$= 0.0665\text{ J}$

E_A - E_B

$0.06424 - 0.06650 = -0.00230\text{ J}$

In an ideal experiment, E_A (initial total mechanical energy) should be equal to E_B (final total mechanical energy) because the total mechanical energy in the system is conserved, and therefore the values in the last column in Table 2 should be zero. However, you will probably find that the two values, E_A and E_B, are not exactly equal because of the experimental uncertainties involved. Next, perform the overlap test to check whether the two values **E_A and E_B are equal within their experimental uncertainties** (Notes to Students: Section 5).

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How to compare E_A and E_B using the overlap test: Select one trial to perform sample calculations and the overlap test. First we must calculate the absolute uncertainties, δE_A and δE_B.

Absolute uncertainty, δE_A

Since $E_A = U_A + K_A$, you can write, $\delta E_A = \delta U_A + \delta K_A$. Use the δU_A value found in the last part. Calculate δK_A using the following equation. Use t_A, v_A and E_A values that you selected from Table 2.

$$K_A = \frac{1}{2} m v_A^2 = \frac{1}{2} m \left(\frac{l}{t_A}\right)^2, \quad \delta K_A = \left(\frac{\delta m}{m} + 2\frac{\delta l}{l} + 2\frac{\delta t_A}{t_A}\right) K_A \tag{16}$$

$$= \left(\frac{0.0001}{0.2179} + 2\frac{0.010}{0.125} + 2\frac{0.001}{0.222}\right) 0.03454$$

$$= 0.00585 \text{ J}$$

$$\delta E_A = \delta U_A + \delta K_A = 0.00666 + 0.00585 = 0.01251 \text{ J}$$

The total initial mechanical energy = $E_A \pm \delta E_A$

Record the above values in **intermediate value** format.

$$E_A + \delta E_A = 0.06424 + 0.01251 = 0.0768 \text{ J}$$

Absolute uncertainty, δE_B

$E_B = U_B + K_B = K_B$ and therefore, $\delta E_B = \delta K_B$.

We can use the Equation (17) to calculate δK_B. However, we have to use t_B and v_B instead of t_A and v_A. Use t_B, v_B, and E_B values obtained for the trial you selected.

$$K_B = \frac{1}{2} m v_B^2 = \frac{1}{2} m \left(\frac{l}{t_B}\right)^2, \quad \delta K_B = \left(\frac{\delta m}{m} + 2\frac{\delta l}{l} + 2\frac{\delta t_B}{t_B}\right) K_B \tag{17}$$

$$\delta E_B = \delta U_B + \delta K_B = \delta K_B$$

The total initial energy = $E_B \pm \delta E_B$

Record the above values in **intermediate value** format.

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OVERLAP TEST: Finally, compare E_A and E_B using the overlap test. An example of overlap test is given in Notes to Students: Section 5.1. Use the intermediate values of δE_A and δE_B (i.e., 2 significant figures).

$$|E_A - E_B| \leq (\delta E_A + \delta E_B) \tag{18}$$

Is the test successful? If $E_A = E_B$, what does this mean? Write your answers in the space above.

EXPERIMENT 3: WORK DONE BY AN EXTERNALLY APPLIED FORCE

DATA AND CALCULATIONS

Table 3: Work Done by an External Force

(initial) t_A (s)	(final) t_B (s)	$v_A = \ell/t_A$ (m/s)	$v_B = \ell/t_B$ (m/s)	$E_A = {}^1\!/\!_2 m v_A^2$ (Initial energy) (Joules)	$E_B = {}^1\!/\!_2 m v_B^2 + mgh$ (Final energy) (Joules)	$W_{ext} = E_B$ - E_A (Joules) Experimental	$W_{ext} = F\ L$ (Joules) Theoretical
0.180	0.101						
0.182	0.101						
0.181	0.101						

Average W_{ext}:

Theoretical work done, W_{ext}:
The force **F** is the tension in the string attached to the car. An analysis of the system used in this experiment (Figure 5) shows that the magnitude of **F** is given by Equation (19) where: $\sin\theta = H_{avg} / D$, m = mass of the air car, m_1 = 30 g = mass attached to the string. Calculate the theoretical value of F using Equation (19), and then W_{ext} using Equation (20). F is an intermediate value. Think about how many significant figures you should keep in the answer.

$$F = \frac{m\, m_1}{m + m_1} g\ (1 + \sin\theta) \tag{19}$$

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$$W_{ext} = F\,L \tag{20}$$

Compare the average W_{ext} from Table 3 to the theoretical W_{ext} above using the percent difference test. Is the test successful? If $W_{ext,\,experimental} = W_{ext,\,theoretical}$ what does that mean? (See the example in Section 5.2, Notes to Students).

Percent difference test:

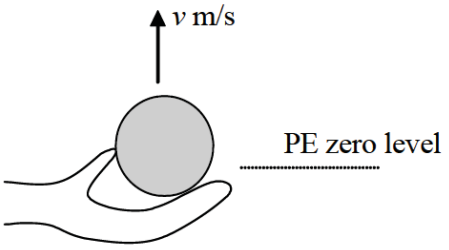
$$\frac{\left|W_{ext,\,experimental} - W_{ext,\,theoretical}\right|}{\left(W_{ext,\,experimental} + W_{ext,\,theoretical}\right)/2} \times 100 < 10\% \tag{21}$$

If $W_{ext,\,experimental} = W_{ext,\,theoretical}$ within 10% error, what does that mean?

Post-Lab Question: Answer the following questions without help from your demonstrator. You may discuss with your lab partner.

1) Suppose you throw a ball of mass m straight up with an initial vertical velocity v (m/s) and catch it later. Answer the following questions assuming there is no air friction or no non-conservative forces acting on the ball. Define potential energy as zero at the level where the ball leaves your hand.

Your answers **must contain only *m* and *v*** and no other variables like height. You can find the answers easily if you think about the conservation of mechanical energy of the system.



(a) What is the kinetic energy of the ball when it was leaving your hand? _____

(b) What is the mechanical energy of the ball when it was leaving your hand? _____

(c) What is the mechanical energy of the ball at the maximum height? _____

(d) What is the mechanical energy of the ball when you catch it later? _____

2) How much work is done by gravity on the hanging mass in Experiment 3 as it drops a height L ?

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CONCLUSIONS AND DISCUSSION

Summarize all your conclusions drawn from your results. Separate your conclusions into different sections rather than writing an essay. Discuss your results. Are they successful? If not, explain why? Your results must support your explanation. Do you have any suggestions to improve any part of your experiment?

Final Mark	